Computational Birdsong Analysis: From Biological Foundations to Current Methods

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Abstract

Birdsong is a key model for studying vocal learning. However, its analysis is hindered by a laborious and largely manual annotation process. With recent advances in technologies like machine learning, and particularly deep learning, computational methods have begun to address this bottleneck. Nevertheless, the field lacks a focused overview that maps current approaches and their limitations. This paper offers the first dedicated survey of computational methods for birdsong analysis. We outline the biological and data foundations of the task, propose a categorization of existing methods, and highlight open challenges. We call on the community to establish shared datasets, benchmarks, and biologically informed models to advance the field.

1. Introduction

Bird vocalizations offer a rich model for studying complex sequential behaviors, with parallels to human speech in its structured organization. They can be divided into two main groups - calls and songs. Songs are longer and more complex. They are typically composed of discrete acoustic units, called syllables, which are separated by silence or amplitude changes. Then, arranged into motifs and song bouts, following complex species-specific rules [BOBB11].

Analyzing birdsong sheds light on the brain mechanisms of vocal behavior across species, offering a valuable comparative model for understanding human speech and language [DK99]. While it lacks semantic content, its hierarchical and rule-governed structure mirrors aspects of the human syntax, making birdsong well-suited to computational modelling and analysis.

These methods, given the complexity and scale of vocal datasets, as well as the limitations of manual annotations, can play an important role in advancing birdsong research, offering scalable and reproducible solutions to a complex task. Therefore, the main objective of this paper is to provide a detailed overview of the current state of computational birdsong analysis, focusing on classification of the existing methods. Moreover, the paper includes brief history of the task and overview of available datasets and evaluation metrics. It also identifies common challenges, and concludes with a call to action for future work.

The rest of the paper is organized as follows: In Section 2, we discuss related work that has addressed similar classifications, either within a more limited scope of the field or in the broader domain of bioacoustics. Section 3 provides a biological background of the birdsong, including anatomy, structure of the song and the most commonly studied bird species. Section 4 includes datasets and metrics. In Section 5 we present our classification of current computational methods for birdsong analysis, and in Section 6 we conclude the paper with a call to action for future research directions in the field.

2. Related Work

To the best of our knowledge, there is currently no published survey or overview paper that systematically compares computational methods for birdsong analysis or gives a detailed overview of the current state of research in the field. Only a few isolated studies approach such comparisons within a limited scope. Individual techniques are widely explored, but there haven't been much progress in doing comparative studies.

Nicholson [DN16] conducts a small-scale evaluation of three machine learning classifiers applied to the song of the Bengalese finch. Bonet et al. [BSAP*21] explore a broader acoustic event detection framework, comparing feature extraction and classification algorithms, with birdsong included as one of several test scenarios. Dong and Jia [DJ20] offer one of the most comprehensive overviews in the domain of automatic bird species recognition from environmental audio. Their work surveys a wide range of methods, from noise removal and bird call detection to feature extraction and classification, and reviews datasets and software tools. However, their focus remains on species-level identification and does not address the structural or syntactic properties of birdsong. More recently, Koch et al. [KMR24] present AVN, a deep learning pipeline for birdsong analysis, which includes a performance comparison against existing methods for motif similarity and developmental stage classification. Although the primary focus of their work is on proposing a novel model, the inclusion of such comparisons underlines the value of systematic benchmarking and highlights the scarcity of broader comparative studies in the field. Therefore, this paper aims to address the lack of focused comparative analysis on computational methods for birdsong.

3. Biological Background

In this section, we provide a brief biological overview of the birdsong, focusing on its anatomical basis, structural organization, and species-specific characteristics. This background lays the foundation for understanding the challenges and motivations behind computational approaches to birdsong analysis.

3.1. Vocal Anatomy and Learning

Songbirds, which belong to the suborder *Passeri* within the order *Passeriformes*, are known for their ability to produce complex vocalizations called birdsong. These sounds are generated by the syrinx - a highly specialized vocal organ located at the junction of the bronchi and trachea. The syrinx allows birds to produce rapid and diverse sound patterns, and its structure and control are the focus of ongoing biological research [DK99].

Birdsong is typically seasonal and is mainly used for two social functions - attracting mates and defending territory. It is distinct from bird calls, which are shorter, simpler vocalizations often used to signal alarm or maintain contact within a flock [CS03]. Unlike calls, birdsong is learned behavior that shares important developmental features with human speech [ZZZ*23]. The process of song learning happens in two main phases. First, during the sensory phase, a young bird listens to and memorizes the song of an adult tutor [Zan96], [MKG*15]. Then, in the sensorimotor phase, it gradually learns to reproduce the song using auditory feedback to refine its vocal output. This learning process depends on both innate mechanisms and experience, and it provides a valuable model for studying vocal learning and brain plasticity [Moo09].

3.2. Birdsong Structure

The basic units of the birdsong are notes (or elements) - brief sounds that combine to form syllables. Syllables are often grouped into repeated patterns called motifs. A sequence of motifs, produced in a particular order, forms what is known as a song bout [BOBB11]. These bouts may vary in complexity and length depending on the species. In some species, the order of elements in the song is mostly consistent, while in others it can be flexible and richly variable, resembling the syntactic variation found in human language [ZZZ*23]. Despite these similarities, a fundamental difference with the human language is the lack of compositional semantics, meaning that birds do not combine song elements to convey novel meanings [BOBB11].

3.3. Model Species in Birdsong Research

A few songbird species have become central models in computational birdsong analysis, primarily due to their distinct vocal characteristics and well-documented learning behaviors. Figure 1 summarizes the bird species examined across the reviewed studies, marking their presence within each paper.

Zebra finch, *Taeniopygia guttata*, is the most widely used species, particularly valued for its simple, stereotyped song consisting of a single motif repeated consistently. This makes it ideal for studying the acquisition and neural encoding of fixed song patterns [Moo09], [FS10].

Bengalese finch, *Lonchura striata domestica*, in contrast, produces songs with variable syntax and branching sequences, offering a useful model for investigating sequence generation and probabilistic structure in vocal output [Oka04], [JK11].

Canary, *Serinus canaria*, is known for its highly complex, long, and hierarchically structured songs. They change seasonally and exhibit adult plasticity, making the bird a prime subject for studying song variability and long-term vocal modulation [LC02], [MIKG13].

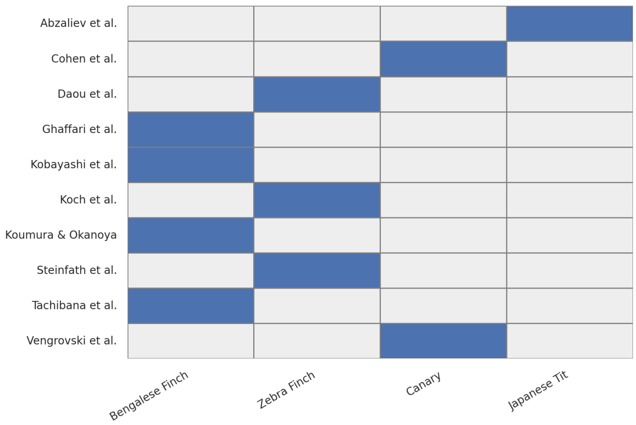

Figure 1: *Bird species used in reviewed papers: Abzaliev et al. [AISM24], Cohen et al. [CNS*22], Daou et al. [DJWB12], Ghaffari et al. [GJD23], Kobayashi et al. [KMT*25], Koch et al. [KMR24], Koumura & Okanoya [KO16], Steinfath et al. [SPMR*21], Tachibana et al. [TOO14], Vengrovski et al. [VHVBG25]*

3.4. Traditional Analytical Methods

Traditional birdsong analysis relies heavily on manual practices rooted in auditory perception and visual inspection. Researchers usually record vocalizations and generate spectrograms, which are then annotated by hand to identify syllables, phrases, or song types [CS03], [MS04]. Identification and classification of these song elements is guided by the knowledge of a human expert, using tools and software to measure pitch, duration, and inter-syllable intervals [Kro82]. Usually, syllables can be grouped based on shape and structure.

Naturally, these techniques introduce significant limitations - they are time-consuming, lack scalability, and are subject to inter-observer variability. Cross-species comparisons are especially difficult due to inconsistencies in annotation protocols and acoustic variability. Therefore, these constraints of manual annotation, along with the increasing availability of large-scale audio recordings, are a motivation for the development of automated and semi-automated computational methods. The aim is to reduce human labor, improve reproducibility, and scale across datasets and species.

4. Datasets and Metrics

Datasets are an essential part of the research and development, because new methods and algorithms need to be tested and compared. In this section, we provide a brief overview of datasets, used for computational birdsong detection and analysis. Moreover, we discuss the current state of metrics for evaluating models.

Datasets vary in terms of size, species coverage, recording conditions, and annotation quality. Large-scale repositories such as *Xeno-canto* [VPPJ17] and *NIPS4Bplus* [MBP*18] provide extensive collections of bird vocalizations from thousands of species across the globe. These crowd-sourced recordings have been instrumental in species recognition and large-scale classification tasks. However, their limited annotations, typically at the recording or species level without precise vocal element boundaries, makes them less suitable for tasks such as syllable segmentation or song structure analysis. In contrast, most research focused on syllable-level segmentation, classification, and sequence modeling relies on smaller, species-specific datasets that provide detailed manual annotations. These include datasets of Bengalese finch songs [KO16], [DN16], canary vocalizations [CNS*22], and zebra finch recordings [SPMR*21], [DJWB12]. Such datasets typically contain recordings from a limited number of individuals, which are annotated at the syllable or motif level, making them well-suited for training and evaluating models.

Metrics. The evaluation of computational models for birdsong analysis currently lacks standardized metrics, complicating direct comparisons across studies. Different approaches often rely on a mix of metrics borrowed from related fields such as speech processing or general audio event detection. Cohen et al. [CNS*22] assess the performance of their model, TweetyNet, in two ways. They use syllable error rate, which is an analogue to the word error rate - the standard metric for automatic speech recognition. In addition to that, they compare the predicted annotations of their model against key findings from previous studies of Bengalese finches and against statistical models of song syntax [CNS*22]. Ghaffari et al. [GJD23] and Steinfath et al. [SPMR*21], for example, employ traditional classification metrics, including F1 score, precision and recall, to evaluate the performance of their models. Another metric, used in the work of Vengrovski et al. [VHVBG25] and Koch et al. [KMR24], is the V-measure statistic [RH07], which evaluates clustering quality by balancing homogeneity (each cluster contains only members of a single class) and completeness (all members of a class are assigned to the same cluster). This diversity of metrics poses challenges for benchmarking and reproducibility of results.

5. Computational Methods

In this section, we propose a categorization scheme for the different computational approaches applied to birdsong analysis. While we acknowledge that many research works may not fall exactly into a single category and that overlaps are common, we believe that this taxonomy can help bringing clarity and structure to this rapidly evolving field. We organize the methods into three dimensions:

Methodological paradigms: Distinguishing between approaches relying on traditional machine learning techniques versus those that utilize deep learning architectures.

Input modalities: Differentiating whether the models operate directly on raw audio signals or on transformed representations such as spectrogram images.

Learning frameworks: Grouping methods according to their supervision strategy - supervised, semi-supervised, or unsupervised learning.

5.1. Methodological Paradigms: Traditional vs Deep Learning

5.1.1. Traditional

Early work in birdsong analysis relies on classical machine learning techniques with hand-crafted acoustic features. Daou et al. [DJWB12] develop rule-based tools for analyzing syntactic structure in song sequences, which require expert-designed rules and are difficult to generalize across species.

Tachibana et al. [TOO14] apply supervised Support Vector Machine (SVM) to semi-automatically classify Bengalese finch syllables. They use a total of 532 acoustical features, which are calculated for each extracted syllable, managing to achieve high accuracy with relatively little training data - one-minute recordings, which equals to approximately 400 syllables. However, this approach is sensitive to noise, and as the previous one, it requires that all syllables are inspected and labeled by an expert.

Abzaliev et al. [AISM24] focus mainly on proposing a taxonomy to classify the phonemes in the vocalizations of Japanese tit, *Parus minor*. Therefore, in order to do this, they also explore machine learning methods to automatically detect and classify files. They work with both spectrogram-based audio classification and audio converted to discrete tokens. They apply few-shot machine learning techniques to classify phonemes, but the model achieves an accuracy of only 13.9%.

5.1.2. Deep Learning

With the advent of deep learning, more flexible and scalable models have emerged. Koumura and Okanoya [KO16] introduce a hybrid model of multi-layered neural network and a hidden Markov model for syllable classification and boundary detection, improving robustness to sequence variability. The authors discuss a common problem with hand-crafted audio features - it is not clear if these features, for example, mel-frequency cepstral coefficients (MFCC), are actually suitable for birdsong analysis or suitable for species different than the ones used in the experiments. Therefore, they suggest that methods should be able to extract good features automatically. They claim that their method achieves this, because deep neural networks usually learn good feature representations from the data without manual feature engineering. Similar to previous studies, two major problems are the need for labeled data for training and the existing background noise in the recordings.

In their work, Cohen et al. [CNS*22] propose TweetyNet, a neural network architecture that directly segments spectrograms of birdsong, significantly reducing the need for manual annotations. They describe their architecture as consisting of two main structural elements - a convolutional block and a bidirectional Long Short-Term Memory (LSTM) layer [HS97].

DAS - DeepAudioSegmenter, is another method, introduced by

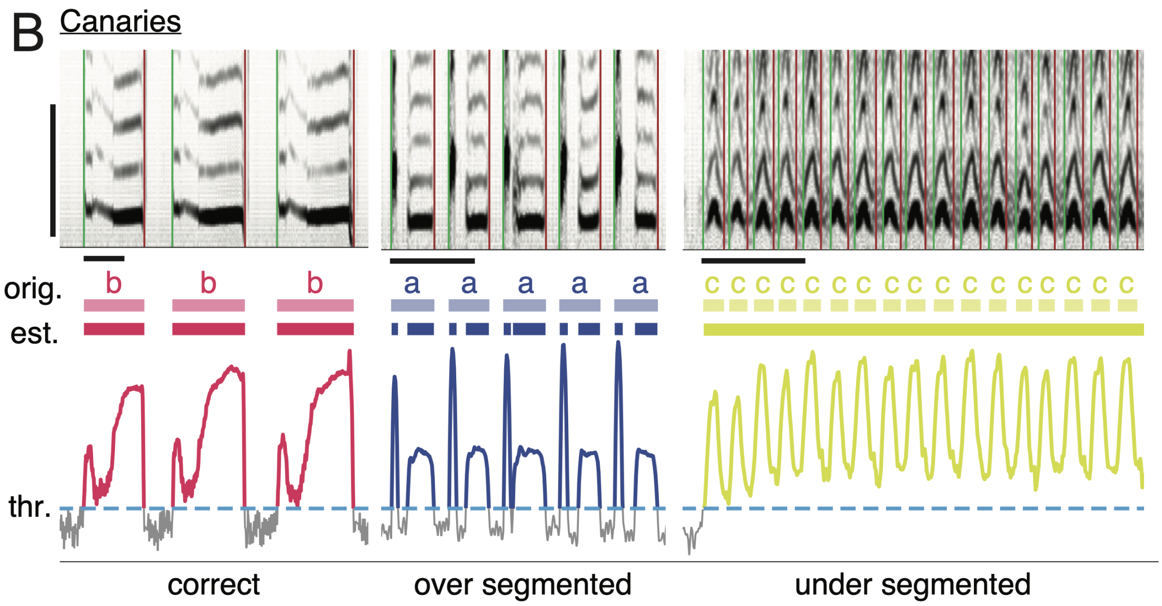

Figure 2: *Wrong segmentation of syllables using the amplitude threshold method as shown in the work of Cohen et al. [CNS*22].*

Steinfath et al. [SPMR*21], for automatic annotation of acoustic signals, requiring little manual annotations. It works with single or multi-channel audio data as input. Central to its architecture is the use of Temporal Convolutional Network [LFV*17]. Their method performs well across different species.

Ghaffari et al. [GJD23] propose a semi-supervised method for the syllable segmentation step of birdsong analysis, which should be the first step in a fully automated framework. Using the Mean Teacher framework [TV17], they train a student model guided by a teacher network on unlabeled data, combining convolutional layers with a bidirectional LSTM. To improve robustness and reduce bias, they apply data augmentation and confidence threshold technique to produce pseudo-labels.

Koch et al. [KMR24] introduce anther deep learning pipeline, AVN, for zebra finch song annotation, which functions across multiple colonies without retraining. AVN can automatically segment and label syllables. Furthermore, it can extract interpretable syntax, timing, and acoustic features, predict developmental stage, and calculate a novel song imitation score.

TweetyBERT, is an example of a self-supervised Transformer model [VSP*17], presented by Vengrovski et al. [VHVBG25]. It is trained on spectrograms of canary songs and its goal is to predict masked audio segments. The model is able to parse units - notes, syllables, phrases, and to capture their acoustic and temporal structure.

Most recently, Kobayashi et al. [KMT*25] introduce FinchGPT, a generative Transformer language model trained on a textualized corpus of Bengalese finch song. Evaluated on next-syllable prediction, FinchGPT significantly outperforms traditional baselines — Markov chains, RNNs, and LSTMs, achieving lower cross-entropy and demonstrating superior modeling of long-range dependencies. Analysis of its attention patterns further confirms its ability to capture hierarchical and syntactic structures in birdsong, paralleling its success in human language modeling. To the best of our knowledge, the authors are the first ones trying to leverage generative models for birdsong analysis.

Despite these advances, several key challenges persist. Traditional machine learning approaches often require extensive manual effort for feature engineering. This hinders the performance of the models, when they are faced with inter-individual or inter-species variations. Deep learning models offer substantial advantages in terms of automatic feature extraction and scalability. Thus, they show promising performance across species and tasks. However, they usually need a lot of annotated data and are prone to overfitting when trained on small, species-specific datasets. Additionally, many deep models behave as black boxes, making it difficult to interpret learned representations in biologically meaningful ways. Training also remains computationally expensive and may require domain-specific tuning to perform well across different species or song types. To summarize, currently, deep learning methods offer flexibility and great potential for automation, but the challenge to produce robust, generalizable, and also interpretable models for birdsong analysis still remains.

5.2. Data Modalities: Raw Audio vs Spectrograms

Studies on computational methods for birdsong analysis can also be separated into two main groups based on the type of data input. The first one is working with raw audio data, while the other group utilizes spectrograms.

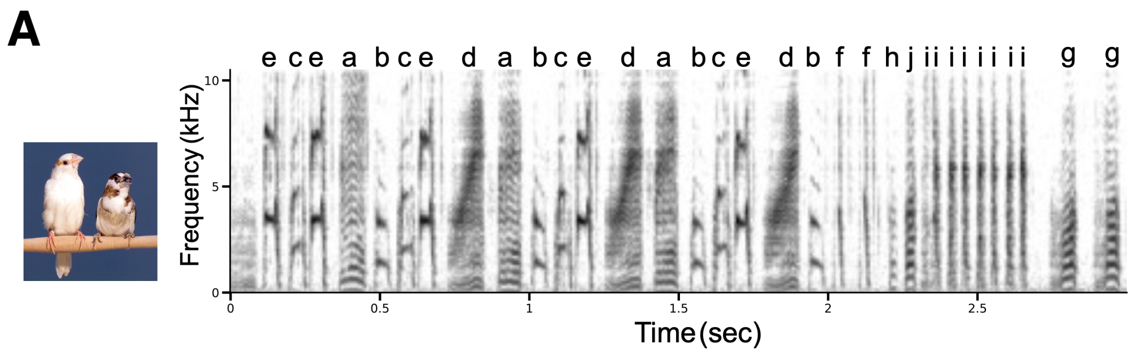

Figure 3: *Spectrogram of Bengalese finch song with syllables labelled with arbitrary alphabet symbols [KMT*25].*

5.2.1. Raw Audio

When working with audio data, a widely used approach is to utilize an algorithm that segments the audio in samples, based on an amplitude threshold. By setting a specific threshold, the goal of the algorithm is to find continuous time points above that threshold. However, this method cannot be reliable in all cases, especially for birdsong, like the canary song, where the amplitude varies greatly, and there is no threshold that can reliably segment all of the syllables as shown in Figure 2 [CNS*22]. Another common problem of raw audio input is the mentioned in Section 5.1 problem of hard-engineered audio features. It is not known if these are good representation of the birdsong data. The earlier works of Daou et al. [DJWB12] and Tachibana et al. [TOO14] are examples for this problem. One study that works with raw input data without engineered features is the work of Steinfath et al., which utilizes temporal convolutional networks to learn sound features that are task specific [SPMR*21].

5.2.2. Spectrograms

Utilizing spectrograms, which are visual representations of the frequencies of an audio signal with time, is compatible with convolutional architecture, and does not require the segmentation of audio data. The way this method works, as described by the authors of TweetyNet, is by taking a window from a spectrogram as input. Then, the produced output is a label for each of the time bins of this spectrogram [CNS*22]. Figure 3 shows an examples of a spectrogram of the Bengalese finch song with labelled syllables [KMT*25]. However, the spectrograms as input also have drawbacks. For example, the fact that they are unsuitable for short signals and in some cases the pre-processing transform needs hand-tuning [SPMR*21]. Furthermore, in the case of complex signals, the recurrent units used in models like TweetyNet, can be difficult to train, as well as slow to run [SPMR*21]. Another aspect of using spectrograms, that Vengrovski et al. [VHVBG25] discuss, is the fact that they utilize only high-resolution spectrograms and parameters that are usually used in the field. Therefore, it would be interesting if the performance of the model is good only when using specific parameters or it could achieve equally good results with a wider range of parameter settings.

5.3. Learning Frameworks: Supervised, Semi-supervised, and Unsupervised

5.3.1. Supervised

Supervised learning remains the most common paradigm with models trained on annotated datasets. Koumura and Okanoya [KO16] and Tachibana et al. [TOO14] exemplify this approach, relying on labeled syllables for classification and boundary detection. FinchGPT [KMT*25] is also an example of a supervised generative model.

5.3.2. Semi-supervised and Unsupervised

However, acquiring high-quality annotations is time-consuming and costly. As a result, semi-supervised and unsupervised methods are becoming increasingly important. An interesting approach is the work of Ghaffari et al. [GJD23]. They use a deep semi-supervised framework to leverage both labeled and unlabeled data. It is trained on only $\sim$ 20 labeled songs, and the model is able to achieve near-expert-level performance on hours of recordings and to generalize across individuals within a species. This demonstrates that semi-supervised learning can significantly reduce annotation effort while maintaining high segmentation accuracy. Similarly, Steinfath et al. [SPMR*21] show that their model, DeepAudioSegmenter, can be extended to weakly supervised and unsupervised scenarios. With TweetyBERT, Vengrovski et al. [VHVBG25] also propose a self-supervised transformer neural network, which manages to cluster elements of a birdsong only from a masked prediction task. The authors report results, which are very close to the ones from human annotations.

6. Conclusion and Future Work

This paper attempts to provide an overview of the current landscape in computational birdsong analysis. It aims to guide both newcomers and experienced researchers in the field through the biological foundations, available datasets, and methodological advances. Moreover, we attempt to examine current work, and categorize it according to methodological paradigms, input modalities, and learning frameworks. To the best of our knowledge, this is the first overview that is exclusively dedicated to computational methods for birdsong analysis, which reflects a significant gap in the field. We believe that by summarizing existing approaches, our work can provide a reference point for future research and help for a more coordinated development.

Despite recent progress, key obstacles persist: the scarcity of large, high-quality annotated datasets, the lack of standardized evaluation protocols, and the difficulty of developing models that generalize across individuals, species, and acoustic conditions. At the same time, emerging methodologies, especially those leveraging deep learning, semi-supervised and self-supervised learning, can offer promising solutions to reduce annotation effort and improve model scalability. Therefore, we also want to use this paper as a call-to-action upon the research community to focus on these missing foundations. This could be achieved by developing robust benchmarking practices, creating open and reusable datasets, and collaborations that integrate biology and technological advances. Addressing these challenges will not only accelerate technical innovation in birdsong analysis, but also contribute to our broader understanding of vocal learning, sequential behavior, and animal communication.

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